



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
21-08-2021

JEE(Main + Advanced) : NURTURE COURSE PHASE-TEPS, TOPS, I, I(A), I(B), II(A) & NURTURE LIVE-I, I(B), II(A)

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

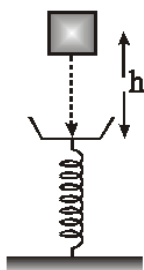
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

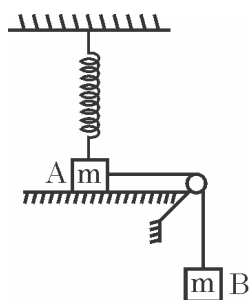
1. A block of mass m initially at rest is dropped from a height h on to a spring of force constant k . The maximum compression in the spring is x then :



- (A) $mgh = \frac{1}{2}kx^2$ (B) $mg(h+x) = \frac{1}{2}kx^2$
- (C) $mgh = \frac{1}{2}k(x+h)^2$ (D) $mg(h+x) = \frac{1}{2}k(x+h)^2$
2. The potential energy U for a force field $\vec{F}$ is such that $U = -kxy$, where k is constant. Then
- (A) $\vec{F} = kx\hat{i} + ky\hat{j}$ (B) $\vec{F} = -ky\hat{i} - kx\hat{j}$
- (C) $\vec{F} = ky\hat{i} + kx\hat{j}$ (D) None of these

Space for Rough Work

3. Under the action of a force, a 2kg body moves such that its position x as a function of time is given by $x = \frac{t^3}{3}$ where x is in metre and t in seconds. The work done by force in first two seconds is
- (A) 1.6 J (B) 16 J (C) 160 J (D) 1600 J
4. The potential energy function for the two atoms is expressed as $U(r) = a/r^{12} - b/r^6$. The equilibrium distance between them is-
- (A) $(a/b)^{1/6}$ (B) $(2a/b)^{1/6}$ (C) $(b/a)^{1/6}$ (D) $(b/2a)^{1/6}$
5. Two blocks A and B, each having a mass of 480 gm connected by a light string passing over a smooth light pulley. The block A is attached to a spring of spring constant 60 N/m whose other end is fixed to a support 40 cm above the horizontal surface as shown. Initially, the spring is vertical and unstretched when the system is released to move. Find the velocity of the block A at the instant it just breaks off the surface below it. (take $g = 10 \text{ m/s}^2$)



- (A) 1.54 m/s (B) 1.44 m/s (C) 1.34 m/s (D) 1.64 m/s
6. A particle of mass ' m ' connected to a string of length ' ℓ ' which is rotating in a vertical circle such that the ratio of the maximum to the minimum tension in the string is 2 : 1 then find out the speed of the particle at the bottom most point.
- (A) $\sqrt{13g\ell}$ (B) $\sqrt{12g\ell}$ (C) $\sqrt{11g\ell}$ (D) $\sqrt{10g\ell}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

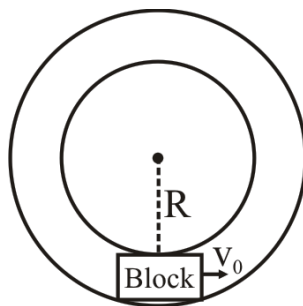
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

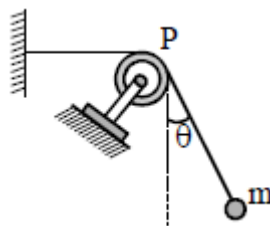
7. Block is free to move inside a vertical fixed smooth tube shown in the figure. Choose correct option(s)



- (A) Block will oscillate when $v_0 = \sqrt{2gR}$
- (B) Block will complete vertical circular motion when $v_0 = \sqrt{4.5gR}$
- (C) Block will complete vertical circular motion when $v_0 = \sqrt{6gR}$
- (D) Block will complete vertical circular motion when $v_0 = \sqrt{2gR}$

Space for Rough Work

8. A bob of mass $m = 10 \text{ kg}$ hangs by a light inextensible string which is connected with a rigid support after passing over a smooth pulley P. If the bob is released from an angular position $\theta = 30^\circ$, at the given position.



- (A) The force exerted by the string on the pulley is $50\sqrt{3}\text{N}$
 (B) The net horizontal force exerted by the string on the mass is $25\sqrt{3}\text{N}$
 (C) The force exerted by the string on the wall is $50\sqrt{3}\text{N}$
 (D) Acceleration of the bob at given instant is zero.
9. Potential energy function along x-axis in a certain force field is given as

$$U(x) = \frac{x^3}{3} - \frac{5}{2}x^2 + 6x - 6$$

For the given force field, choose the correct option(s).

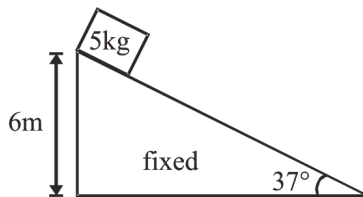
- (A) the points of equilibrium are $x=2$ and $x=3$
 (B) the point $x=2$ is a point of unstable equilibrium
 (C) the point $x=3$ is a point of stable equilibrium
 (D) there exists no point of neutral equilibrium

Space for Rough Work

10. Choose the correct alternative(s):

- (A) When a conservative force does positive work, the potential energy of the system decreases.
- (B) Work done by friction on a body always results in a loss of its kinetic energy.
- (C) Work done in the motion of a body over a closed loop is zero for every force in nature.
- (D) Work done by friction on a body may increase its kinetic energy.

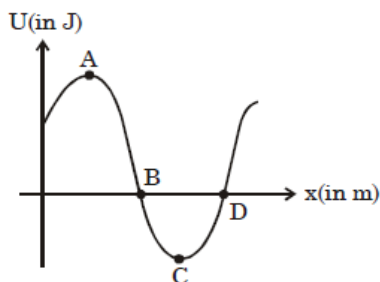
11. A block is released from rest on an fixed incline plane from a height 6m above the ground as shown in figure where coefficient of friction between block and incline plane is 0.5. Choose the correct option(s) for motion of block from shown position to the bottom of incline :-



- (A) Change in potential energy is -300 J .
- (B) Work done by normal is 400 J .
- (C) Work done by friction is -200 J .
- (D) Final velocity of block when it just reaches to bottom of incline plane is $2\sqrt{10} \text{ m/s}$.

Space for Rough Work

12. In the given figure the variation of potential energy of a particle of mass 'm' is represented w.r.t. its x-coordinate. The particle can move under the effect of this conservative force along the x-axis. Select the **CORRECT** alternatives :



- (A) At point B, the acceleration of particle is along the positive x-axis.
 (B) At point C, the acceleration of particle is along the positive x-axis.
 (C) At point A and C, particle is in unstable and stable equilibrium respectively.
 (D) At point B and D, particle is in stable and unstable equilibrium respectively.

Space for Rough Work

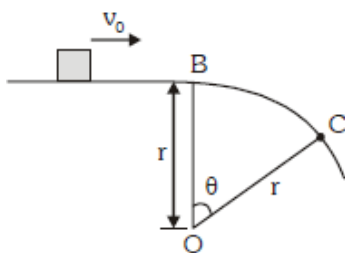
SECTION-II : (Maximum Marks: 24)

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- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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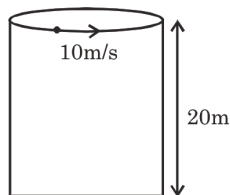
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Zero Marks : 0 In all other cases.

1. A small frictionless block slides with velocity $0.5\sqrt{gr}$ on the surface as shown in the figure. The block leaves the surface at point C. The angle $\theta = \cos^{-1}\frac{x}{8}$ then find x :

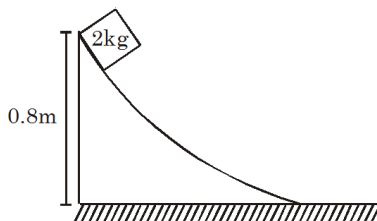


2. A particle is thrown horizontally with speed 10 m/s along the rim of a smooth fixed cylinder of height 20m. Taking $g = 10 \text{ m/s}^2$, the time taken by the particle to reach the bottom assuming it to be always in contact with the cylinder is (in sec).

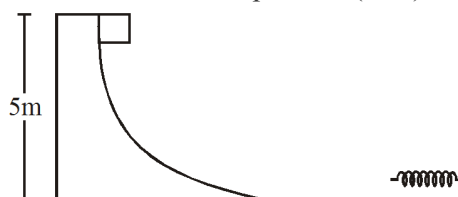


Space for Rough Work

3. When 2kg mass is released from top of a fixed incline plane, then final velocity of 2kg block (in m/s) when it reaches on ground is
(Given : All surfaces are smooth)

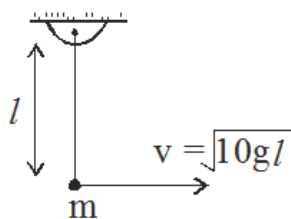


4. Figure shows a smooth curved track terminating in a smooth horizontal part. A spring of spring constant 25 N/m is attached at one end to a wedge fixed rigidly with the horizontal part. A 1kg mass is released from rest at a height of 5 m on the curved track. Find the maximum compression (in m) of the spring. ($g = 10 \text{ m/s}^2$)



Space for Rough Work

5. A ball tied to a massless thread undergoes vertical circular motion as shown. Its speed at the lowest point of motion is $\sqrt{10gl}$. Find 'X' if its speed at highest point is $\sqrt{Xgl}$.



6. A certain system has potential energy given by the function $U(x) = -8x^2 + x^4$. What is the x coordinate of the point where the equilibrium is unstable?

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

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Negative Marks : -1 In all other cases

1. Consider the hypothetical situation where the azimuthal quantum number (ℓ) takes values 0, 1, 2, $n + 1$, where n is the principal quantum number. In the element with atomic number ($z = 36$) the last electron enter in which subshell ? (Rule for filling of electron are fixed).
(A) 3f (B) 4p (C) 2d (D) 3g
2. Which of the following process can be endothermic.
(I) $O + e^- \rightarrow O^-$
(II) $O^- + e^- \rightarrow O^{2-}$
(III) $Ne + e^- \rightarrow Ne^-$
(IV) $N + e^- \rightarrow N^-$
(A) I, II (B) II, III (C) II, III, IV (D) I, II, III, IV
3. From 2 mg calcium, 1.2×10^{19} atoms are removed. The number of g-atoms of calcium left is ($Ca = 40$, $N_A = 6 \times 10^{23}$)
(A) 5×10^{-5} (B) 2×10^{-5} (C) 3×10^{-5} (D) 5×10^{-6}

Space for Rough Work

4. 0.8 gm of BeCl_2 added to water forming 500 mL solution. 20 mL of this solution is taken and diluted 10 folds to form final solution. Find moles of Cl^- ions in 2 mL of final solution :
(Atomic weight Be = 9, Cl = 35.5)
- (A) 2×10^{-6} (B) 4×10^{-6} (C) 8×10^{-6} (D) 1×10^{-6}
5. 50 ml of a gas was burnt with 70 ml of oxygen gas. After reaction the remaining gases occupying 95 ml when passed through KOH gave a contraction of 50 ml. Find the possible gas.
- (A) CH_4 (B) C_2H_4 (C) C_2H_2 (D) CO
6. The correct order of oxidation state of nitrogen in NO, N_2O , N_2O_4 & N_2O_3 is :
- (A) $\text{N}_2\text{O}_4 < \text{N}_2\text{O}_3 < \text{NO} < \text{N}_2\text{O}$ (B) $\text{N}_2\text{O} < \text{NO} < \text{N}_2\text{O}_3 < \text{N}_2\text{O}_4$
(C) $\text{NO} < \text{N}_2\text{O} < \text{N}_2\text{O}_3 < \text{N}_2\text{O}_4$ (D) $\text{N}_2\text{O} < \text{NO} < \text{N}_2\text{O}_4 < \text{N}_2\text{O}_3$
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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7. Which of the following is/are the incorrect match for atom of elements ?

(A) $[\text{Rn}] 6d^2 7s^2 \rightarrow$ d-block, 4th group

(B) $[\text{Xe}] 4f^{14} 5d^1 6s^2 \rightarrow$ d-block, 3rd group

(C) $[\text{Xe}] 4f^7 6s^2 \rightarrow$ f-block, 9th group

(D) $[\text{Kr}] 4d^{10} \rightarrow$ d-block, 12th group

8. Select the correct comparison(s) of boiling point :

(A) $\text{H}_2\text{O} > \text{H}_2\text{S}$

(B) $\text{HF} < \text{HCl}$

(C) $\text{NH}_3 > \text{PH}_3$

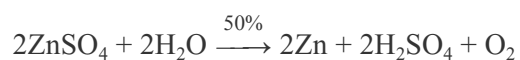
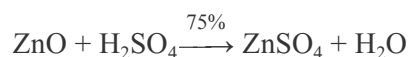
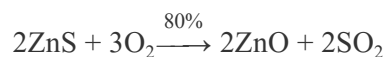
(D) $\text{BCl}_3 < \text{BI}_3$

Space for Rough Work

9. 20 ml of a gaseous compound containing nitrogen and hydrogen only, is passed through a series of electric spark, by which the volume becomes 40 ml. When the resulting gases are passed through water, the volume becomes 30 ml. Now, 50 ml oxygen gas is mixed in the resulting gases and fired. If the final volume becomes 50 ml, the correct information(s) is/are
(All the volumes are measured at the same temperature and pressure). N_2 and H_2 gases are completely insoluble in water)
- (A) The gaseous compound is NH_3
- (B) The gaseous compound is N_2H_4
- (C) Only 50% of the gaseous compound was decomposed into N_2 and H_2
- (D) 100 % of the gaseous compound was decomposed into N_2 and H_2
10. Identify the correct statement(s) among the following :- [$N_A = 6.02 \times 10^{23}$]
- (A) 0.5 gm-atom of oxygen contains $\frac{N_A}{4}$ oxygen molecules
- (B) Total electrons present in 0.5 g-atom of oxygen is $4N_A$
- (C) Mass of 3.01×10^{23} molecules of ammonia is 8.5 g
- (D) Number of atoms present in 11.2 litres of chlorine at 1 atm & 273K is 6.02×10^{23}

Space for Rough Work

11. Given



If 15 mol of ZnS reacts with excess of reactants then which is/are correct statements ?

- (A) 15 mol of Zn formed (B) 2.25 mol of O₂ gas released
 (C) 18 mol of H₂SO₄ formed (D) 9 mol of ZnSO₄ formed

12. Which of the following statements is/are correct?

- (A) NH₂⁺ shows sp² hybridisation whereas NH₂⁻ shows sp³ hybridisation
 (B) Al(OH)₄⁻ has a regular tetrahedral geometry
 (C) sp³ hybridized orbital has equal s- and p- character
 (D) Hybridized orbitals always form σ - bonds and π bonds both

 Space for Rough Work

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-
1. Calculate the number of σ bond in given compound $C_3N_3(N_3)_3$.
 2. Find number of oxyacid which have peroxy linkage in its structure ,
 $H_2S_2O_7$, H_2SO_5 , $H_4P_2O_8$, HNO_3 , H_3PO_5 , H_3AsO_4 , H_2SeO_4 , $H_4P_4O_{12}$
 3. If from 3 moles $MgSO_4 \cdot 7H_2O$, all the 'O' atoms are taken out and converted into ozone (O_3), then find the number of moles of O_3 formed.
 4. 12 gm of an alkaline earth metal reacts with 8 gm of oxygen gas to form its oxide. The atomic weight of metal is (in gram/mole)-
 5. 24 g of carbon (graphite) is completely reacted with 38.4 gm of oxygen and form CO and CO_2 gases at STP, then the ratio of moles of CO_2 & CO formed is x : 1. The value of 'x' is :-
 6. Total number of degeneracies possible in 4th excited state of Li^{+2} ion
-

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

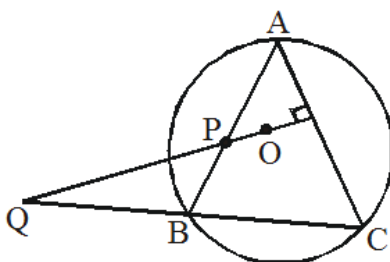
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1. In a $\triangle ABC$, if $(a + b + c)(a + b - c)(b + c - a)(c + a - b) = a^2b^2\lambda$, then -
 (A) $4 < \lambda \leq 6$ (B) $6 < \lambda \leq 8$ (C) $0 < \lambda \leq 4$ (D) $9 < \lambda < 12$
2. If two sides of an acute angled triangle are 6 & 7, then number of possible integral values of length of third side is-
 (A) 3 (B) 4 (C) 6 (D) 9
3. In the adjoining figure, O is the circumcentre of triangle ABC. The perpendicular bisector of AC meets AB at P and CB produced at Q. Then $\angle PQB : \angle PBO$ is-



- (A) 1 : 1 (B) 2 : 3 (C) 3 : 2 (D) 4 : 5

Space for Rough Work

4. Let α, β, γ be the altitudes on the sides a, b, c respectively of a ΔABC & α, β, γ are the roots of the equation $x^3 - px^2 + 44x - 20 = 0$. Inradius of the ΔABC is (with usual notations) -
- (A) $\frac{21}{11}$ (B) $\frac{22}{11}$ (C) $\frac{5}{11}$ (D) $\frac{10}{11}$
5. If $D_r = \begin{vmatrix} r & 2021 & 2022 \\ 11r^2 & 3r & r \\ r & \frac{1}{r} & \frac{1}{r^2} \end{vmatrix}$, then $\sum_{r=1}^5 D_r$ is equal to-
- (A) 0 (B) 155 (C) 2013 (D) -2012
6. Number of solutions of equation $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \cdot \tan 2x} = 1$ in $[-2\pi, 2\pi]$ is equal to-
- (A) 2 (B) 4 (C) 8 (D) 0

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. If the sides of a triangle are in A.P. with common difference 1 and whose circumradius is $\frac{8}{\sqrt{15}}$, then which of the following can be side(s) of a triangle

(A) 2 (B) 3 (C) 4 (D) 5

8. Which of the following is/are true

(A) $\sin(225^\circ) = -\frac{1}{\sqrt{2}}$

(B) $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{8}$

(C) $(1 + \tan 3^\circ)(1 + \tan 42^\circ) = 2$

(D) $\frac{\sin 5\theta + \sin \theta}{\cos 5\theta + \cos \theta} = \tan 3\theta$ (wherever is defined)

Space for Rough Work

9. In ΔABC , sides $a = 8$, $b = 7$ and $A - B = 2\tan^{-1}\left(\frac{1}{\sqrt{23}}\right)$ and its area is $\frac{105\sqrt{x}}{y}$ (where x & y are prime numbers), then the value of $(x + y)$ is divisible by -
 (A) 2 (B) 3 (C) 4 (D) 6
10. The value of the expression $\log_{10}(\tan 6^\circ) + \log_{10}(\tan 12^\circ) + \log_{10}(\tan 18^\circ) + \dots + \log_{10}(\tan 84^\circ)$ is
 (A) a whole number (B) an irrational number
 (C) a negative integer (D) a rational number which is an integer
11. If equation $ax^2 + bx + a = 0$ where $a, b \in \mathbb{N}$ and $x^3 + 2x^2 - 2x - 1 = 0$ have two roots common, then value of $(2a + 3b)$ is not equal to :-
 (A) 5 (B) 11 (C) 13 (D) 17
12. If the equation $2(1 + a^2) = \sin 2\theta + 2a(\sin \theta + \cos \theta)$ has real solution, then which of the following statements is (are) true ?
 (A) sum of all possible values of 'a' is zero
 (B) 'a' can take only two real values.
 (C) Number of values of θ satisfying the equation in $[0, 4\pi]$ are 4.
 (D) Number of values of θ satisfying the equation in $[0, 4\pi]$ are 2.

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\frac{9}{2}$, 6 and $\frac{15}{2}$ are length of medians of ΔABC then area of ΔABC is-
2. If P is the value of $\text{antilog}_{\sqrt[5]{7}} 10$, $Q = 7^{\log_7 6} + (81)^{\log_9 (\frac{5}{2})}$ and R is the number of positive integers having characteristic 2 when base of logarithm is 8, then the value of $\frac{QR}{10P}$ is
3. In ΔABC with usual notations $\frac{\cos A}{a} = \frac{2 \cos B}{b} = \frac{3 \cos C}{c}$ and $\frac{b}{c} \sin 2A = \frac{x\sqrt{y}}{z}$ (where x,z are prime numbers and $y \in \mathbb{N}$), then the value of $\frac{(x+z)}{y}$ is
4. If $\begin{vmatrix} 1 & \cos \alpha - \sin \alpha & \cos \alpha + \sin \alpha \\ 1 & \cos \beta - \sin \beta & \cos \beta + \sin \beta \\ 1 & \cos \gamma - \sin \gamma & \cos \gamma + \sin \gamma \end{vmatrix} = k \begin{vmatrix} 1 & \cos \alpha & \sin \alpha \\ 1 & \cos \beta & \sin \beta \\ 1 & \cos \gamma & \sin \gamma \end{vmatrix}$, (where $\cos \alpha \neq \pm \sin \alpha$, $\cos \beta \neq \pm \sin \beta$, $\cos \gamma \neq \pm \sin \gamma$) then the value of 'k' is
5. 3 natural numbers x, y, z are in G.P. & x, y^2, z are in A.P., then the minimum value of $(x + 2y + z)$ is equal to
6. Let $N = \frac{\overbrace{999\dots 9}^{16 \text{ digits}} - \overbrace{5555\dots 5}^{8 \text{ digits}}}{\overbrace{9000\dots 04}^{9 \text{ digits}}}$, then sum of digits in N is

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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